Book 1 — Lecture 1 — Interlude 1 — Exercise 4

Magnitudes, Dot Product, and Angle

Prompt. Let $\mathbf{A} = (A_x, A_y, A_z) = (2, -3, 1)$ and $\mathbf{B} = (B_x, B_y, B_z) = (-4, -3, 2)$. Compute the magnitudes of $\mathbf{A}$ and $\mathbf{B}$, their dot product, and the angle between them.

Computation

$$\begin{aligned}
\|\mathbf{A}\| &= \sqrt{2^2 + (-3)^2 + 1^2} = \sqrt{14} \approx 3.741657, \\
\|\mathbf{B}\| &= \sqrt{(-4)^2 + (-3)^2 + 2^2} = \sqrt{29} \approx 5.385165, \\
\mathbf{A} \cdot \mathbf{B} &= 2(-4) + (-3)(-3) + 1(2) = 3.
\end{aligned}$$

Here $\|\cdot\|$ denotes vector magnitude[1] and $\mathbf{A} \cdot \mathbf{B}$ denotes dot product[2].

For the angle θ[3] between vectors:

$$\cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{3}{\sqrt{14}\sqrt{29}} = \frac{3}{\sqrt{406}}.$$

Hence,

$$\theta = \arccos\left(\frac{3}{\sqrt{406}}\right) \approx 1.421388 \text{ rad} \approx 81.437539^\circ.$$

Result. $\|\mathbf{A}\| = \sqrt{14}$, $\|\mathbf{B}\| = \sqrt{29}$, $\mathbf{A} \cdot \mathbf{B} = 3$, and the angle is $\theta \approx 81.44^\circ$.

Example. Since $\mathbf{A} \cdot \mathbf{B} > 0$, the angle is acute ($< 90^\circ$), matching the computed value 81.44°.

[1] Euclidean norm, computed as square root of the sum of squared components.

[2] For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$.

[3] By convention, θ often denotes an angle between vectors in geometry and trigonometry.